CEFEPIME

Antipseudomonal

Neurotoxicity in renal failure

CHEMICAL NAME (6R,7R)-7-[[(2Z)-2-(2-amino-1,3-thiazol-4-yl)-2-methoxyiminoacetyl]amino]-3-[(1-methylpyrrolidin-1-ium-1-yl)methyl]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylate (hydrochloride)

CLASS Fourth-generation cephalosporin

TRADE NAMES Maxipime · Cepimax · Cefepime HCl injection

THERAPEUTIC USES / INDICATIONS

- Febrile neutropenia (empirical monotherapy).
- Hospital-acquired and ventilator-associated pneumonia.
- Complicated UTI and pyelonephritis; intra-abdominal infection (with metronidazole).
- Severe skin and soft-tissue infections.

ADMINISTRATION

- **IV** infusion over 30 min (or extended infusion); **IM** for mild UTI.
- Adjust dose according to CrCl.

DOSAGES

- **Adult:** 1–2 g every 8–12 h; febrile neutropenia 2 g every 8 h.
- **Children (>2 months):** 50 mg/kg every 8–12 h (max 2 g/dose).
- **CrCl 30–60:** 2 g every 12–24 h; **11–29:** 1–2 g every 24 h; lower below.

PHARMACOKINETICS

- **Onset:** immediate IV. Good tissue and CSF penetration.
- Minimal metabolism.
- Excreted unchanged in urine. **t½:** 2 h.

PHARMACODYNAMICS

- Inhibits **cell-wall synthesis** → bactericidal.
- Broad spectrum: *Pseudomonas*, Enterobacterales (incl. AmpC producers), and good Gram-positive cover (streptococci, MSSA).
- More stable to chromosomal β-lactamases than 3rd generation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea.
- Rash, headache.
- Phlebitis.

ADVERSE EFFECTS (SERIOUS)

- Neurotoxicity: confusion, myoclonus, non-convulsive status epilepticus, encephalopathy (especially in renal failure/elderly).
- Anaphylaxis.
- *C. difficile* colitis.
- Positive Coombs test.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to cefepime, cephalosporins or severe penicillin allergy.
- **Caution:** renal impairment, elderly, seizure disorders.

KEY INTERACTIONS

- **Aminoglycosides, loop diuretics:** ↑ nephrotoxicity/ototoxicity.
- **Warfarin:** ↑ INR.

NURSING MANAGEMENT

- Check renal function and adjust dose before giving.
- Monitor mental status; report confusion or twitching.
- Monitor temperature, cultures, CBC (neutropenic patients).
- Monitor IV site and diarrhoea.
- Neutropenic precautions.

PATIENT EDUCATION

- Report confusion, jerking movements, rash or severe diarrhoea.

KEY POINTS

- 4th-gen cephalosporin — Pseudomonas + Gram-positive cover.
- Neurotoxicity if not renally adjusted.
- Febrile neutropenia 2 g every 8 h.
- Monitor mental status in elderly.